

Ryan Chan

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Education

University of Toronto

May 2027

Bachelor of Applied Science in Computer Engineering, Minor in Artificial Intelligence & Business, Dean's List Scholar

Work Experience

Bloomberg

June 2026 - Aug 2026

Software Engineering Intern, Risk Performance & Platform Front Office Integrations

New York, NY

- Built a context-aware TypeScript/Rapid UI Bloomberg Terminal workflow enabling sell-side trader clients to launch and manage risk reports directly from their active books; shipped from local development through beta to production
- Partnered directly with Product Managers, sell-side trader clients, and cross-functional teams across time zones to gather requirements and translate workflows into technical specs
- Integrated the workflow with comdb2 services to pull user permissions, enabled risk types, and historical risk reports
- Investigated and resolved client-reported report issues, tracing failures across a distributed systems report orchestration pipeline (Python microservices) using Humio and Splunk logs to identify and ship root-cause fixes

Scotiabank Global Capital Markets

May 2025 - Apr 2026

Quantitative Trading Developer Co-op, Electronic Execution Services & Algorithmic Trading

Toronto, ON

- Embedded in the front office on the trading floor alongside traders daily, gathering requirements, learning workflows, and translating them into production tools they used and tested directly
- Developed and deployed a real-time trade monitoring and reporting system (Angular, FastAPI) with automated distribution of client-facing research and reports
- Built a production Python pipeline (Pandas, NumPy, scikit-learn) to model intraday time-series data for 6,000+ equities, classifying and clustering based on liquidity buckets, improving VWAP algorithm execution accuracy 8%
- Built a real-time Java/FIX Protocol handler ingesting trades from exchanges and brokers into internal systems, and SQL pipelines (Oracle, SQL Server) reconciling and computing analytics over millions of trades downstream

Ada

Sep 2024 - Dec 2024

Software Engineering Intern, Growth & Performance

Toronto, ON

- Built a self-serve onboarding flow (React, Redux, MongoDB) that let customers train and deploy their own AI agents for customer service, cutting time to first deployment from 63 days to 7 days
- Ran feature-flagged A/B tests and PostHog analytics on onboarding flows, pairing behavioral data with direct customer interviews to expand AI-agent adoption and capabilities
- Implemented document re-ranking in a RAG pipeline, improving LLM retrieval precision by 4%
- Built an internal tool (Python, AWS S3, Github Actions) converting 100+ JSON LLM regression tests into version-controlled, human-readable format, making model evaluation accessible across the team

Projects

SimplyMail | Electron, TypeScript, SQLite, Next.js

Apr 2025 - Aug 2025

- Architected a local-first desktop email client supporting 20K+ messages with event-driven IPC architecture
- Implemented Gmail sync using incremental APIs, background workers, and caching for efficient data updates

Ryko | Typescript, Python, PostgreSQL, AWS, LangChain

Apr 2024 - Aug 2024

- Built a RAG system with dynamic retrieval across vector stores and web data (LangChain, LangGraph)
- Developed a document ingestion pipeline (AWS Lambda, ECR) to process unstructured data into vector embeddings
- Built 40+ REST APIs and designed PostgreSQL schema from scratch to support scalable retrieval and querying

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, C, C++, Java

AI & LLM: RAG, LangChain, LangGraph, LlamaIndex, Vector stores, LLM evaluation, MCP

Fullstack: React, Angular, Next.js, Node.js, FastAPI, Express, REST APIs, TailwindCSS

Data & Infrastructure: PostgreSQL, Oracle, SQL, MongoDB, Redis, SQLite, AWS, GCP, Docker, Humio, Splunk